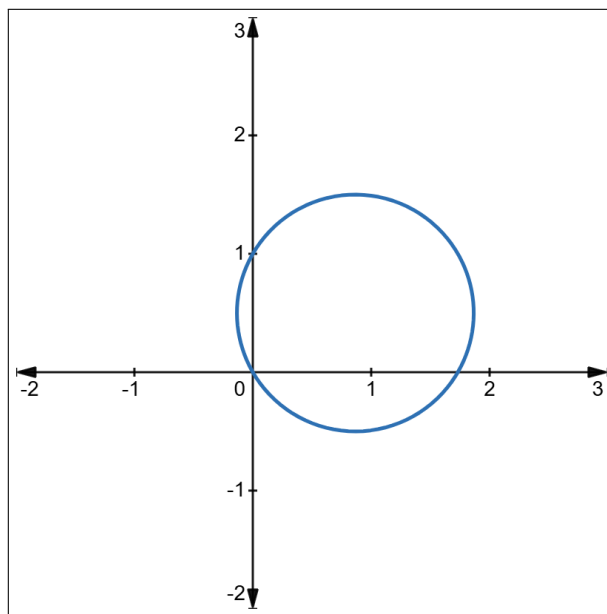


B5 past final worksheet

1. (Fall 2024, #9) Find the area of the segment of the circle with polar equation

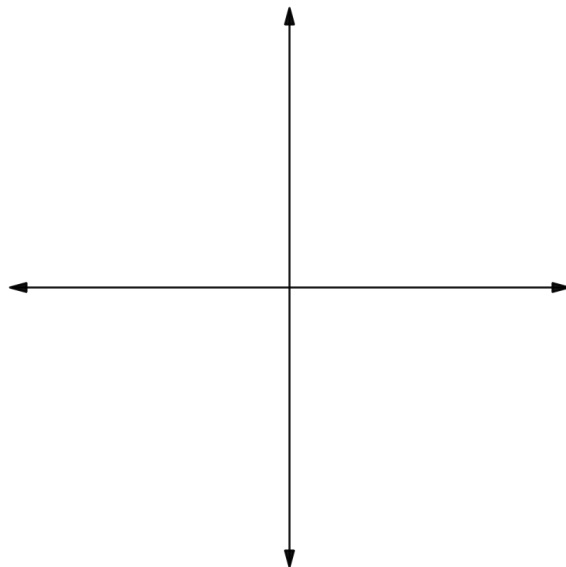
$$r(\theta) = \sqrt{3} \cos \theta + \sin \theta; \quad \theta \in [-\pi/3, 2\pi/3)$$

below the polar axis (in the figure below, the polar axis coincides with the positive x -semiaxis and the pole with the Cartesian origin).



2. (Spring 2023, #6) Consider the polar curve $r = 1 - \sin(\theta)$.

(a) Sketch the curve on the axes below.

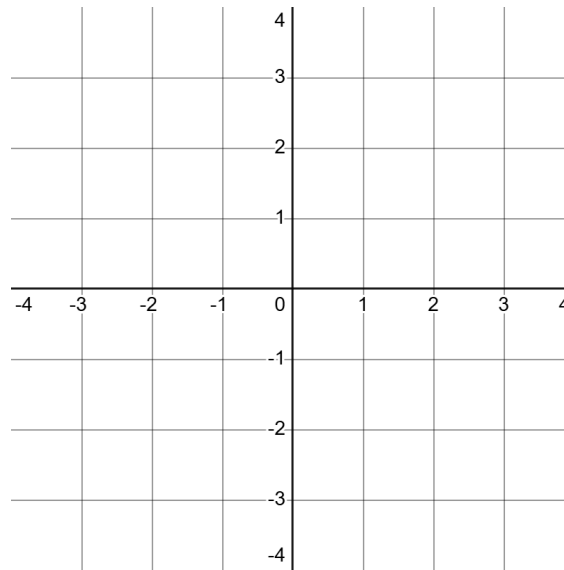


(b) Find the area enclosed by the curve.

3. (Fall 2019, #8) Consider the equation

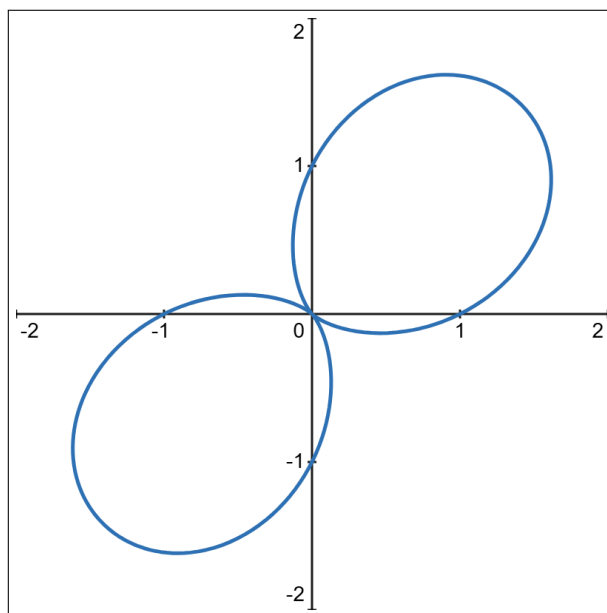
$$r = \sin(4\theta)$$

- (a) In the figure below, draw the polar curve $r = \sin(4\theta)$ for all $0 \leq \theta \leq 2\pi$.



- (b) Find the area enclosed by the curve, in the first quadrant. (The first quadrant is the region where $x \geq 0$ and $y \geq 0$.)
- (c) Find the slope of the tangent line to the curve when $\theta = \pi/4$.
- (d) Write an integral that computes the arc length of this curve where $0 \leq \theta \leq \pi/4$. You do NOT have to evaluate this integral.

4. (Spring 2018, #7) Consider the curve (pictured below) given in polar coordinates by $r = 1 + \sin(2\theta)$.



Find the area of one of the loops pictured above.

5. (Fall 2017, #6) Find the area of the shaded region below. The equations of the curves C_1 and C_2 are given in polar coordinates.

